

Ensuring U.S. Dominance in AI for Biology

Andrew Hickl, CTO, Allen Institute

March 14, 2025

AI is transforming biology, medicine, and drug discovery, creating unprecedented opportunities for scientific breakthroughs. But the global race for dominance in AI-driven biosciences is accelerating — China is investing billions to own this space.¹ Without swift, strategic action, the U.S. risks falling behind in AI-powered biomedical innovation, weakening national security, and losing control over the future of medicine.

The worst-case scenario is a future where critical breakthroughs in medicine and biotechnology are controlled by geopolitical rivals, leaving the U.S. dependent, vulnerable, and unable to protect the health and security of its citizens.

To dominate this space, the U.S. must focus on targeted investment, regulatory flexibility, and public-private partnerships that fuel innovation while protecting national security. We must take action to ensure that scientific advancements flourish in an environment that prioritizes speed and security.

True leadership here means the U.S. sets global standards, drives breakthroughs that transform global health, and captures enormous economic value—positioning America at the forefront of a new era in biotechnology and AI.

We recommend the Government adopt the following strategic priorities and actions to ensure U.S. dominance in AI-driven biology. Immediate, decisive steps will safeguard national security, accelerate groundbreaking discoveries, and secure our global leadership in biotechnology.

- **Deter Threats and Protect Biosecurity:** Securing America's future starts with ensuring AI-powered biotechnology can't be weaponized against us.
- **Create Sovereign AI and Data Resources:** America must control its AI and data resources independently, ensuring our researchers and innovators drive discovery without external interference or dependence.
- **Make American science faster, smarter, and more efficient with AI:** By removing regulatory bottlenecks and setting world-class standards for data, we position American science to innovate rapidly, using AI to drive new discoveries first.
- **Develop American Talent in AI and Biology:** America's advantage rests in its people. Investing in our next generation of talent ensures the U.S. remains unrivaled in AI-powered bioscience innovation.

The recommendations outlined below offer practical, concrete steps to ensure the United States maintains a dominant position in AI-driven biology, protecting our national interests and securing our technological leadership for decades to come.

Deter Threats and Protect Biosecurity

1. Strengthen AI-Enabled Biosecurity to Prevent Emerging Biothreats

AI is accelerating breakthroughs in drug development, vaccine design, and bio-manufacturing, but it also makes it easier for adversaries like China, Russia, and non-state actors to develop engineered biothreats, synthetic

¹ In the past 12 months, China has established [a 60 billion yuan \(approximately \\$8.2 billion\) AI investment fund](#) to bolster its capabilities in this sector. Additionally, Shanghai has launched [a 100 billion yuan \(\\$13.79 billion\) fund](#) to enhance its integrated circuit, biomedicine, and AI industries.

pathogens, and novel toxins. We can't afford to let AI-driven tools enable bad actors to design and synthesize dangerous biological agents that put the American people at risk.

The U.S. must act decisively to prevent AI from being exploited to create bioweapons while ensuring we remain at the forefront of AI-driven bioscience.

1.1. Mandate AI Risk Assessments for Synthetic Biology: All U.S. AI-driven protein engineering, gene synthesis, and pathogen modeling should undergo risk screening before execution to ensure reasonable oversight.

1.2. Create a National AI Biosecurity Task Force: We should establish a cross-agency task force (OSTP, NIH, DARPA, and DOD) to monitor AI applications in biology, ensuring AI-powered synthetic biology techniques/tools cannot be exploited for developing biothreats or toxins.

1.3. Secure DNA Synthesis & Pathogen Data Against AI Exploitation: The U.S. should implement AI-driven sequence screening for DNA synthesis providers to block unauthorized orders of genetic material related to high-risk pathogens like smallpox or SARS-CoV-2.

Create Sovereign AI and Data Resources

2. Take Back Control: Build Public AI Models for Biomedical Breakthroughs

The growing dominance of private-sector AI models in biomedical research raises concerns about access, transparency, and neutrality. Many of these models are trained on proprietary datasets, limiting scientific reproducibility and public oversight. Additionally, privately developed AI models reflect the priorities of their creators, which can influence healthcare decisions, drug discovery, and medical research in ways that may not align with broader public interests. By funding open, government-backed AI models, the U.S. can safeguard scientific objectivity, expand AI accessibility, and secure American leadership in biomedical AI research.

2.1 Create Open, Government-Backed AI Models for Biosciences: To make sure that AI-driven medical advancements remain both accessible and grounded in rigorous science, the U.S. should establish a federally backed AI research program (NIH, NSF, DOE, OSTP) dedicated to developing large-scale foundational AI models for biology and medicine. A federally funded alternative to private-sector models will ensure that AI-powered drug discovery, precision medicine, and genomics remain accessible to all without restrictive licensing barriers.

2.2 Ensure AI Models Reflect Scientific Objectivity, Not Corporate Priorities: Publicly developed models will be transparent, auditable, and designed for scientific integrity, mitigating (or avoiding) sources of potential bias that arise when AI development is controlled solely by large tech firms.

3. Establish AI-First Standards for Biological Data Collection

The United States can gain a competitive advantage over its adversaries if its scientific institutions operate in a way that ensures interoperability/compatibility of our datasets. Interoperability and data standardization are essential to accelerating AI-driven breakthroughs in biosciences. However, the lack of consistent metadata standards across research institutions slows AI model training and hinders cross-institutional collaboration.

3.1 Standardize AI-Compatible Metadata and Labeling for Biological Experiments: The NIH and NSF should require that all federally funded biological research follows standardized data labeling and metadata practices optimized for AI training. The private sector should play a key role in shaping these standards, ensuring that data collected in academic labs, hospitals, and industry research centers is structured for seamless AI integration. Clarity around AI-ready data standards will speed up innovation, allowing AI models to be trained on diverse, high-quality datasets without the friction of incompatible formats.

3.2 Require AI-Optimized Experimental Design for High-Throughput Data Collection: Funding agencies should prioritize research projects that integrate AI-ready experimental designs, ensuring that biological datasets are created with AI applications in mind.

4. Build Scalable AI Infrastructure for Biomedical Research

The U.S. must dramatically scale its AI computing infrastructure to stay competitive in AI-driven biosciences. Without dedicated, high-performance AI computing resources, U.S. researchers will be outpaced by China and other nations aggressively funding AI-driven life sciences.

By leveraging the resources and infrastructure from public-private partnerships such as the Stargate Project, the U.S. can establish dedicated AI supercomputing clusters and secure cloud environments tailored for biomedical research. This would provide researchers with the necessary computational power and data security to advance AI-driven biosciences.

4.1 Expand AI Supercomputing for Bioscience: Establish dedicated AI-optimized computing clusters for biomedical research, ensuring that academic and government researchers have the same access to high-performance computing (HPC) resources as private industry.

4.2 Develop Secure AI Cloud Infrastructure for Biomedical Data: Create a federal AI-cloud initiative that ensures biomedical AI research can be conducted securely at scale, while protecting sensitive genomic and health data from foreign cyber threats.

Make American science faster, smarter, and more efficient with AI

5. Set the global standard for FAIR data to accelerate American discoveries and leadership.

AI-driven discoveries should fuel American scientific leadership, not be locked behind proprietary walls. Open, standardized datasets enable faster, more reproducible science, allowing the U.S. to set the global standard for AI-driven biosciences. However, openness must be balanced with national security considerations, ensuring that adversaries do not exploit U.S.-funded research.

5.1 Incentivize FAIR Data Standards for Federally Funded AI Research: All federally funded AI research should align with FAIR (Findable, Accessible, Interoperable, Reusable) principles, ensuring that biological datasets and AI models remain accessible for scientific use while protecting sensitive data.

To encourage adoption without unnecessary regulatory burdens, the U.S. should:

- **Tie Federal Research Grants to FAIR Compliance:** Require NIH, NSF, and DOE grantees to demonstrate clear FAIR data management plans in grant applications, ensuring research outputs are structured for AI-readiness and broader use.
- **Create Competitive Funding Streams for FAIR-Compliant Research:** Establish new funding incentives for institutions and companies that prioritize transparent, well-structured biological datasets that accelerate AI model training and reproducibility.
- **Establish Tiered Data-Sharing Models:** Develop a structured approach to open science, where lower-risk biological datasets are freely accessible, while sensitive biomedical or national security-related data remains protected with controlled access.

6. Modernize and Streamline AI Regulatory Frameworks for Biomedical Innovation

Current regulatory frameworks are too slow for the pace of AI innovation in biosciences, leaving the U.S. vulnerable to losing its competitive edge. China, for example, is rapidly modernizing its AI approval pathways, leveraging AI to accelerate clinical trials and streamline drug discovery, positioning itself as a global leader in biomedical AI.²

² <https://cset.georgetown.edu/publication/china-and-medical-ai/>

Meanwhile, outdated U.S. regulations — from slow FDA approval cycles to restrictive HIPAA policies—are blocking AI from reaching its full potential in healthcare and life sciences.

If the U.S. does not act, private-sector AI innovation will be stifled, and adversaries will dictate the future of AI-driven medicine. A modern regulatory approach is critical to speeding up AI deployment in clinical settings, ensuring that groundbreaking AI models can be used safely, efficiently, and at scale without unnecessary bureaucratic delays.

6.1 Adopt Risk-Based AI Regulation: Update FDA and NIH regulatory pathways to support/accommodate faster AI-driven discovery cycles, ensuring AI models can be safely and efficiently deployed in drug development and clinical settings.

6.2 Reform HIPAA & 21st Century Cures Act for AI in Healthcare: AI models need large-scale patient data to personalize treatments, but current HIPAA restrictions make it difficult for AI researchers to access and integrate multi-institutional health records.

6.3 Accelerate AI-Based Clinical Trial Design: The FDA and NIH should formalize AI-powered clinical trial guidance, allowing virtual trials and AI-simulated patient cohorts to accelerate drug approvals.

6.4 Reduce IP & Patent Barriers for AI-Generated Discoveries: The USPTO should create clear patent guidance for AI-generated biological discoveries, ensuring AI-assisted innovations in genomics and drug discovery are protectable without unnecessary delays.

7. Support AI-Driven Discovery of New Biological Principles to Improve Human Health

AI is allowing researchers to uncover new biological principles faster than ever before – opening new frontiers in medicine and making what was previously out-of-reach possible. However, China and other adversaries are aggressively investing in AI-powered biological research as well – without the commitment to open science and intellectual property sharing that is hallmark of so much American science. If the U.S. fails to act, we risk ceding leadership in foundational biomedical research, which underpins future advances in drug development, precision medicine, and synthetic biology.

To maintain U.S. dominance in AI-driven biosciences, we must ensure that AI is not just used for analysis and automation but also as a driver of fundamental discovery.

7.1 Create AI-Enabled Virtual Biology Labs: The U.S. should establish federally funded AI-powered simulation environments where researchers can test AI-generated biological theories before conducting costly lab experiments.

7.2 Accelerate AI Hypothesis Validation with Automated Labs: AI should not just propose new hypotheses — it should also be used to validate them, too. By integrating AI-driven robotics and automated lab workflows, researchers can test AI-discovered biological principles at scale, compressing years of wet-lab research into months.

Develop American Talent in AI and Biology

8. Improve AI Workforce Development for Biology and Medicine

If the U.S. does not invest aggressively in training AI-literate biologists, clinicians, and computational scientists, it risks losing leadership in the biosciences altogether. To make this a reality, the U.S. must deepen collaboration between academia, government, and industry. A key opportunity lies in expanding public-private partnerships to train AI-biology experts who can seamlessly transition between research institutions, biotech firms, and AI startups.

8.1 Fund AI-Biology Training Programs for Researchers and Clinicians: The NIH, NSF, and DOE should create dedicated AI training grants for biologists and clinicians, equipping them with the AI expertise needed to lead the next generation of biomedical research. These programs should focus on practical, application-driven AI training, ensuring the U.S. workforce remains competitive in an era where AI is shaping the future of medicine and biotechnology.

8.2 Develop AI-Focused Fellowships and Externships in Biomedical AI: The U.S. should establish federally funded AI-biology fellowships that place computational scientists in biomedical research environments and biologists in AI-driven tech settings. These fellowships should be structured as public-private collaborations, allowing companies, research institutes, and government agencies to co-host fellows, fostering AI expertise across sectors while ensuring the U.S. private sector benefits from top-tier AI talent.